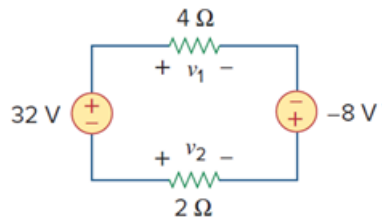


Problem

Find v_1 and v_2 in the circuit of Fig. 2.22.

Figure 2.22:



Step-by-step solution

Step 1 of 4

The circuit is shown in Figure 1.

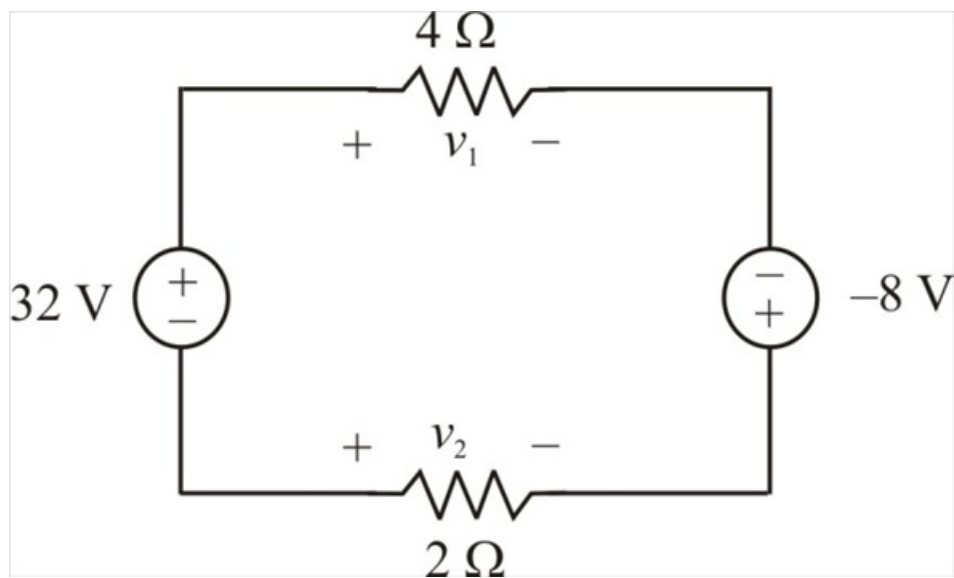


Figure 1

Step 2 of 4

Assume that the current i flows through the loop as shown in Figure 2.

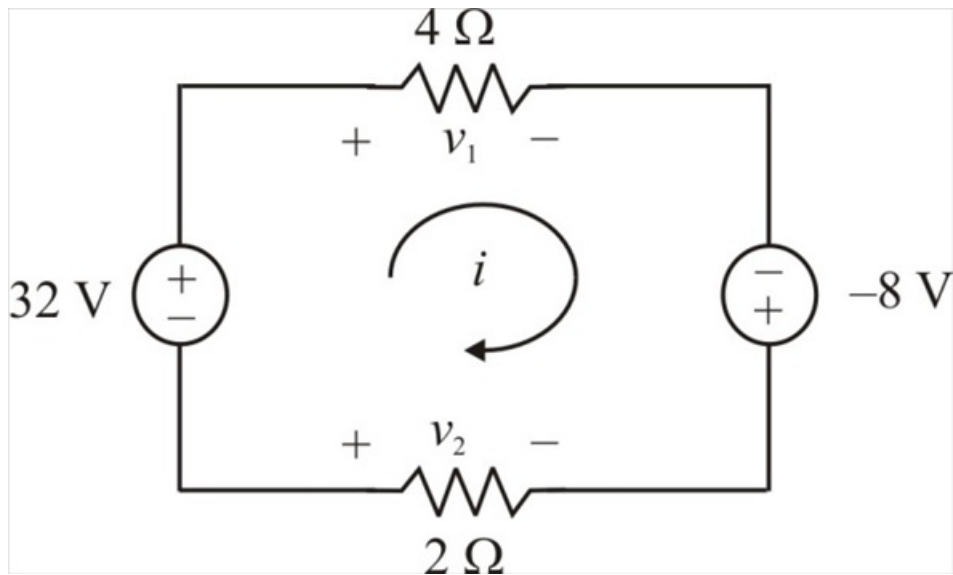


Figure 2

Step 3 of 4

To find v_1 and v_2 , apply Ohm's law and Kirchhoff's voltage law.

From Ohm's law,

$$v_1 = 4i \dots\dots (1)$$

$$v_2 = -2i \dots\dots (2)$$

From Figure 2, apply Kirchhoff's voltage law around the loop.

$$-32 + v_1 - (-8) - v_2 = 0$$

$$-32 + 4i - (-8) - (-2i) = 0$$

$$6i = 24$$

$$i = 4 \text{ A}$$

Substitute 4 for i in $v_1 = 4i$.

$$v_1 = 4i$$

$$= (4)(4)$$

$$= 16 \text{ V}$$

Therefore the voltage v_1 is 16 V.

Step 4 of 4

Substituting the value of i in equation (2).

$$v_2 = -2i$$

$$= (-2)(4)$$

$$= -8 \text{ V}$$

Therefore the voltage v_2 is -8 V.